MIL-P-14536 (Ord)
12 June 1957
SUPERSEDING
PA-PD-22 (Rev 1)
10 May 1954

MILITARY SPECIFICATION

POLYISOBUTYLENE BINDER

1. SCOPE

1.1 Scope. - This specification covers polyisobutylene binder for use in explosive compositions.

1.2 Classification. - Polyisobutylene binder shall be of the following types, as specified (see 6.1):

Type I - containing di-(2-ethylhexyl)-sebacate

Type II - containing di-(2-ethylhexyl)-adipate

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves; Standard, For Testing Purposes
LLL-B-631 - Boxes; Fiber, Corrugated (For Domestic Shipment)

MILITARY

MIL-R-398 - RDX
MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Heavy-Duty
MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition
MIL-D-10692 - Di-(2-ethylhexyl)-sebacate
MIL-P-13298 - Polyisobutylene (For Ordnance Use)
MIL-D-13383 - Di-(2-ethylhexyl)-adipate

FSC 9320

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STANDARDS

FEDERAL

FED. TEST METHOD STD. NO. 791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing

MILITARY

MIL-STD-129 - Marking For Shipment and Storage

DRAWINGS

ORDNANCE CORPS

81-3-169 - Vacuum Stability Test Apparatus

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Physical and chemical properties. - The material shall conform to the following physical and chemical requirements. Compliance with these requirements shall be determined in accordance with the corresponding test procedure specified in section 4 of this specification.

3.1.1 Odor. - The material shall have no odor other than the mild characteristic odor of the lubricating oil.

3.1.2 Volatile matter. - 0.6 percent, maximum.

3.1.3 Composition (volatile-free basis). - The composition of the polyisobutylene binder shall be as specified in table I.

Table I. - Composition

Constituent	Type I	Type II
Polyisobutylene (molecular weight approximately 100,000 to 120,000 - conforming to Specification MIL-P-13298), percent	25.0 ± 0.5	25.0 ± 0.5
Di-(2-ethylhexyl)-sebacate (conforming to Specification MIL-D-10692), percent	59.0 ± 1.0	---
Di-(2-ethylhexyl)-adipate (conforming to Specification MIL-D-13383), percent	---	59.0 ± 1.0
Lubricating oil, Grade 10 (conforming to Specification MIL-L-2104), percent	16.0 ± 1.5	16.0 ± 1.5

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3.1.4 Penetration. - 22.0 ± 3.0 millimeters (mm).

3.1.5 Foreign matter. - 0.5 percent, maximum.

3.1.6 Acid number. - 3.0, maximum.

3.1.7 Unincorporated lumps. - None.

3.1.8 Grit. - No gritty particles shall be retained on a No. 40 U. S. Standard sieve, and not more than 5 gritty particles per 50 grams of the material shall be retained on a No. 60 U. S. Standard sieve, following extraction with benzene (see 4.4.8).

3.1.9 Reactivity. - When tested at 100° C. in contact with RDX, not more than 1 milliliter (ml) of gas shall be liberated (see 4.4.9).

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.

4.1.1 Pilot lot. - The first batch produced shall constitute the pilot lot manufactured under any contract. However, a pilot lot shall not be required when a new contract immediately follows the manufacture of satisfactory material of the same type. A batch shall be defined as that quantity of material which has been subjected to some unit physical or chemical mixing process intended to make a final product substantially uniform.

4.1.2 Production lot. - A production lot shall consist of material from batches subjected to the same processing operation and conditions. The process used to produce a production lot shall be identical to that used to prepare the pilot lot.

4.2 Sampling. - Five 1-pound samples shall be taken at random from each lot. If the lot consists of five batches or less, one sample shall be taken from each batch. Each sample should be placed in an airtight container and labeled to show the name of the material, manufacturer, plant, contract or purchase order number, batch number, and lot number. All acceptance tests shall be made on these samples. However, if by visual observation it becomes apparent during sampling that the batches are not uniform, the inspector may require that additional batches be sampled and tested for compliance with the requirements of this specification.

4.3 Inspection. - Inspection shall be as specified in Specification MIL-C-2550.

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4.4 Tests.

4.4.1 Applicable acceptance tests.

4.4.1.1 For pilot lots. - Tests for acceptance of pilot lots shall consist of all the tests specified herein, and shall be conducted at a Government laboratory designated by the procuring officer.

4.4.1.2 For production lots. - Tests for acceptance of production lots shall consist of all the tests specified herein, with the exception of the reactivity test (see 3.1.9 and 4.4.9).

4.4.2 Volatile matter. - An accurately weighed portion of 7 to 10 grams of the sample shall be transferred to a tared glass dish, 2-1/2 to 3-1/2 inches in diameter and not more than 1 inch deep, and heated in an oven at 100° to 105° C. for 6 hours. The loss in weight of the sample shall be calculated to percentage of volatile matter.

4.4.3 Composition (see 6.2).

4.4.3.1 Polyisobutylene. - An accurately weighed portion of approximately 1.0 gram of the material shall be transferred to a tared 100-ml beaker. A 30-ml portion of a 1:1 ether-acetone solution (see 6.2) shall be added and, with the aid of a glass stirring rod flattened at one end, the sample shall be thoroughly mashed to assure complete penetration of the solvent into the rubbery mass of polyisobutylene. This treatment shall be continued for approximately 15 minutes, and the sample then allowed to remain in contact with the solvent for 1/2 hour with only occasional mashings. The solvent shall be transferred to a tared medium-porosity, sintered-glass crucible using low suction. An additional 30-ml portion of the ether-acetone solution shall be added to the beaker, and allowed to stand for approximately 1/2 hour. The mass of polyisobutylene shall then be transferred to the crucible with the aid of the stirring rod. The beaker shall be rinsed thoroughly inside, and under the lip on the outside, with a stream of the ether-acetone solution, allowing the rinsings to run into the crucible. Suction shall be applied to the crucible until the odor of ether-acetone is no longer detectable. The crucible and contents shall be dried in an oven at 100° to 105° C. for 1/2 hour, cooled in a desiccator, and weighed. The filtrate shall be retained for the determination of di-(2-ethylhexyl)-sebacate or di-(2-ethylhexyl)-adipate, as applicable (see 4.4.3.2 and 4.4.3.3). The polyisobutylene content shall be calculated as follows:

$$\text{Percent polyisobutylene} = \frac{100A}{W}$$

in which:

A = gain in weight of the crucible, in grams
W = weight of sample (volatile-free basis), in grams

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4.4.3.2 Di(2-ethylhexyl)-sebacate (applicable to Type I). - The contents of the filtering flask from the determination of polyisobutylene (see 4.4.3.1) shall be transferred to a 250-ml pyrex iodine flask. The flask shall be placed on a steam bath and the solvent evaporated until the odor of acetone is no longer detectable. The flask shall be removed from the steam bath, cooled, and 25 ml 0.5N alcoholic potassium hydroxide, made from aldehyde-free 95-percent alcohol and carbonate-free potassium hydroxide (see 6.3), shall be added using a calibrated pipette. A flask and air condenser shall be assembled and set into a metal controlled heat transfer plate (see 6.4), to minimize bumping. This assembly shall be placed on a hot plate at low heat. This mixture shall be allowed to reflux gently for two hours. A blank shall be conducted at the same time using the same quantity of 0.5N alcoholic potassium hydroxide. (A few glass beads shall be added to reduce bumping.) The flask shall be removed from the hot plate, cooled to room temperature, and the inside of the condenser carefully washed down with a minimum quantity of absolute alcohol previously made neutral to phenolphthalein with a few drops of 0.1N alkali. The excess alkali shall be titrated with 0.3N hydrochloric acid, using phenolphthalein as the indicator. In case the solution is too highly colored to permit the use of phenolphthalein, the following potentiometric titration shall be made:

The saponified solution shall be transferred quantitatively to a 250-ml beaker, with the aid of approximately 30 ml of 95-percent alcohol. The electrodes of a previously calibrated pH meter shall be inserted into the solution in the beaker, and the solution titrated with 0.3N hydrochloric acid to a pH value of 10.3. The solution shall be stirred continuously during the titration. The end point shall be approached dropwise. The blank shall be titrated in the same manner as the sample. The di-(2-ethylhexyl)-sebacate content shall be calculated as follows:

$$\text{Percent di-(2-ethylhexyl)-sebacate} = \frac{21.321(B-V)N}{W}$$

in which:

B = ml of hydrochloric acid solution used to titrate the blank.
V = ml of hydrochloric acid solution used to titrate the sample.
N = normality of the hydrochloric acid solution.
W = weight of the sample on a volatile-free basis, in grams.

In case of failure or dispute, the potentiometric titration procedure shall be used, taking as the equivalence point the point of greatest change in potential, or pH, per unit of titrant; that is, where Δ pH/Δ ml is a maximum.

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4.4.3.3 Di-(2 ethylhexyl)-adipate (applicable to Type II). - The determination shall be made in the same manner as for di-(2-ethylhexyl)-sebacate (see 4.4.3.2). The di-(2-ethylhexyl)-adipate content shall be calculated as follows:

$$\text{Percent di-(2-ethylhexyl)-adipate} = \frac{18.52(B-V)N}{W}$$

in which:

B = ml of hydrochloric acid solution used to titrate the blank.
V = ml of hydrochloric acid solution used to titrate the sample.
N = normality of hydrochloric acid solution.
W = weight of the sample on a volatile-free basis, in grams.

4.4.3.4 Grade 10 lubricating oil. - The percentage of Grade 10 lubricating oil, on a volatile-free basis, shall be calculated by subtracting from 100 the sum of the percentages of polyisobutylene and di-(2-ethylhexyl)-sebacate, or di-(2-ethylhexyl)-adipate, as applicable, in the material under test.

4.4.4 Penetration. - A portion of the sample shall be transferred to a metal dish approximately 3 inches wide and at least 2 inches deep, so that the surface of the sample is not more than 3 mm from the top of the dish. The dish and contents shall be placed in an oven at 100° to 105° C. until no air bubbles exceeding 2 mm in diameter remain in the gel (overnight at 100° to 105° C. should be sufficient). The dish shall then be placed in an air bath maintained at 25° ± 1° C. for at least 4 hours. The penetration shall then be determined at the center of the dish, and at four points at least 1-1/4 inches from the center of the dish, using the apparatus and procedure specified in Federal Test Method Standard No. 791, Method 311, for unworked penetration of greases. The penetration shall not be determined at any point where there are air bubbles within 3 mm of the surface.

4.4.5 Foreign matter. - An accurately weighed portion of approximately 5 grams of the material shall be transferred to a 400-ml beaker, 250 ml ml of benzene added, and heated on a steam bath with occasional stirring, until the material appears to be completely dissolved. The resulting solution shall be filtered while still hot by means of a tared Gooch crucible having a heavy asbestos mat. The crucible shall be washed 5 times with hot benzene, dried in an oven at 100° to 105° C., cooled in a desiccator, and weighed. The gain in weight of the crucible shall be calculated to percent foreign matter in the sample.

4.4.6 Acid number. - An accurately weighed portion of approximately 5 grams of the sample shall be transferred to a 400-ml beaker. A 75-ml portion of toluene shall be added, and the beaker and contents heated on a

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steam bath, with occasional stirring, until the sample appears to be completely dissolved. A 75-ml portion of alcohol, which has been previously neutralized with 0.1N sodium hydroxide solution using 0.5 ml of phenolphthalein solution as the indicator, shall be added to the solution with constant stirring, then 75 ml of distilled water shall be added. The resulting mixture shall be titrated immediately, using 0.1N sodium hydroxide solution, to a light pink end point. The acid number shall be calculated as follows:

$$\text{Acid number (milligrams of NaOH per gram)} = \frac{40.0\ AN}{W}$$

in which:

A = ml of sodium hydroxide solution used in the titration.
N = normality of the sodium hydroxide solution.
W = weight of sample, in grams.

4.4.7 Unincorporated lumps. - A tray measuring approximately 10 by 10 inches, and having sides approximately 1 inch high, shall be made by folding up the ends of a 12-inch square of stencilboard (see 6.5). A portion of 100 ± 5 grams of the material shall be placed in the center of the tray, and the tray and its contents placed in an oven at 150° ± 5° C. for 4 hours. At the end of the heating period the sample will have spread out into a circular layer approximately 8 inches in diameter. The tray and its contents shall be allowed to cool to room temperature and observation shall then be made to see whether or not any bumps appear on the surface. If any bumps are present they shall be pressed with a finger, and it shall be noted whether they are air bubbles or hard or resilient lumps. If any hard or resilient lumps are present the material shall be considered to contain unincorporated polyisobutylene.

4.4.8 Determination of grit. - A weighed portion of approximately 50 grams of the material shall be transferred to a 2000-ml beaker, 1000 ml of benzene added, and the mixture heated on a steam bath with occasional stirring until the material appears to be completely dissolved. The solution shall be poured into a No. 40 sieve placed over a No. 60 sieve, making sure that all of the insoluble matter is transferred to the sieves. The sieves shall conform to Specification RR-S-366. Any particles retained on the sieves shall be washed with benzene, and the sieves dried. Any particles remaining on each of the sieves shall be counted and examined. It shall be noted if the particles are gritty, as indicated by the persistance of a scratching noise when the material is rubbed with a smooth spatula on a smooth glass slide.

4.4.9 Reactivity (applicable to pilot lot only).

4.4.9.1 Calibration of capillary tubes. - The unit capacity of the capillary tube of the apparatus shown in figure 1 shall be determined as follows: The volume in ml of the 15.5-centimeter (cm) heating tube shall

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be determined by running mercury from a burette into the tube until it is filled to the level at which the ground glass joint of the capillary tube will make contact with the mercury. A 7.0-ml portion of mercury shall be transferred to the cup at the lower end of the capillary tube, the tube clamped in a vertical position, and the height in mm of the mercury column in the capillary tube measured. The length in mm of each of the three parts of the capillary tube shall be measured, and the sum of the three values obtained. The height of the mercury column (approximately 25 mm) determined as outlined above shall be subtracted from the sum of the three values. The capacity of the capillary tube per unit of length (ml per mm) shall be determined as follows: An accurately weighed sample of approximately 10 grams of mercury shall be transferred to the cup at the lower end of the capillary tube. The tube shall be manipulated so that, when the tube is horizontal, mercury is contained in a continuous section of the longest part of the tube. The length of the mercury column shall be measured. This procedure shall be repeated twice with the mercury column located in two other sections of the longest part of the tube. The three lengths so measured shall be averaged, and the unit capacity of the capillary tube calculated as follows:

$$\text{Unit capacity (ml per mm)} = \frac{W}{DL}$$

in which:

W = weight of mercury used, in grams.
L = average length of the mercury column, in mm.
D = density of mercury at temperature of determination.

4.4.9.2 Procedure. - Three heating tubes, of the type used in the apparatus shown in figure 1, are required for this test. The following materials shall be placed in the respective tubes:

Tube No. 1 - 2.5 grams of RDX conforming to Specification MIL-R-398.

Tube No. 2 - 2.5 grams of the polyisobutylene binder under test.

Tube No. 3 - 2.5 grams of RDX and 2.5 grams of the polyisobutylene binder.

The ground glass portion of each tube shall be cleaned with a clean cotton cloth. A light smear of vaseline shall be placed on the jointing ends of three capillary tubes, and by an upward and circular motion the heating tubes shall be connected to the capillary tubes, as shown in figure 1. The apparatus shall be clamped so that the long section of the capillary tube is in a vertical position, and the lower end sits on a solid support. A 7.0-ml portion of mercury shall be transferred to the cup and the lower end of each capillary tube. By means of a vacuum pump connected to the mouth of the mercury reservoir, the system shall be evacuated until the pressure is reduced to approximately 5 mm of mercury. (If the apparatus is tilted forward until the capillary opening of the reservoir is free of mercury, evacuation will be facilitated.) The pump shall be disconnected and the

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total vertical height of the column of mercury in each capillary tube shall be measured. The respective vertical heights of the mercury in the cups shall be measured and subtracted from the mercury column height of the corresponding capillary tube. This value should be within 5 mm of the prevailing barometric pressure; otherwise the assembly shall be reevacuated to the required pressure. The room temperature and the barometric pressure shall be noted. The three tubes and their contents shall be inserted into a constant temperature bath, as shown on Ordnance Corps Drawing 81-3-169, containing a glycerin-in-water solution of approximate specific gravity 1.05. A temperature of $100^{\circ} \pm 0.5^{\circ}$ C. shall be maintained in the heating tubes for 40 hours. The heating tube shall then be removed, supported in a vertical position, and allowed to cool to room temperature. The vertical height of the mercury column shall again be measured and the corresponding vertical height of the mercury in the cup subtracted therefrom. The room temperature and the barometric pressure shall again be noted. The volume of liberated gas, at standard temperature and pressure, shall be calculated as follows:

$$\text{Volume of gas, in ml} = \left[A \times B(C-H)\right]\left[\frac{273P}{760(273+T)}\right] - \left[A \times B(C-H_1)\right]\left[\frac{273P_1}{760(273+T_1)}\right]$$

in which:

A = volume of the heating tube minus the volume of the material added to the tube (assume 1 gram = 1 ml).

B = the unit capacity of the capillary tube, in ml per mm (see 4.4.9.1).

C = total length of the capillary tube minus the vertical height of the mercury in the cup at the end of the test, in mm.

H = vertical height of mercury column in the capillary tube minus vertical height of mercury in the cup at end of test, in mm.

P = barometric pressure at end of test minus "H", in mm.

T = room temperature at end of test, $^{\circ}$C.

H_1 = vertical height of mercury column in the capillary tube minus vertical height of mercury in the cut at beginning of test, in mm.

P_1 = barometric pressure at beginning of test minus "H_1", in mm.

T_1 = room temperature at beginning of test, $^{\circ}$C.

The reactivity of the polyisobutylene binder in contact with RDX shall then be calculated as follows:

$$\text{Reactivity (in ml of gas)} = X - (Y+Z)$$

in which:

X = ml of gas produced by the mixture of RDX and polyisobutylene binder.

Y = ml of gas produced by the RDX alone.

Z = ml of gas produced by the binder alone.

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4.5 Acceptance.

4.5.1 Pilot lot. - The pilot lot shall be considered acceptable provided the test samples comply with all the requirements specified in section 3 of this specification.

4.5.2 Production lot. - The production lot shall be considered acceptable provided the test samples comply with all the requirements specified in section 3 of this specification, with the exception of the reactivity requirement. The procurement activity shall have the option of accepting individual batches from a rejected production lot, if the samples representing those batches comply with all the applicable requirements.

4.6 Resubmission and retest.

4.6.1 Pilot lot. - If the pilot lot is rejected, the contractor may correct the condition causing the production of the defective batch, and present a new batch as a resubmitted pilot lot. The resubmitted lot shall be accepted provided the sample complies with all requirements of this specification.

4.6.2 Production lot. - If a production lot is rejected, the contractor may have the option of having a partial or complete analysis made on samples taken from any or all of the batches in the lot, at no expense to the Government. The contractor may then remove the defective portion of the lot, and resubmit the lot. The resubmitted lot shall be sampled by taking a sample from each batch. The resubmitted lot shall be accepted provided all samples comply with all the applicable requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level C. - Unless otherwise specified polyisobutylene binder shall be packaged in 55-gallon, round, open-top commercial steel drums. The drums will require no over packing.

5.2 Marking. - The containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).

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6.2 Reagent chemicals. - Ethyl ether, acetone, alcohol, and benzene used in performing the composition tests should be of analytical reagent grade (American Chemical Society specifications) (see 4.4.3).

6.3 The method described in Analytical Chemistry, Vol. 21, May 1947, page 359, Englis and Reinschreiber, "Determination of Saponification Numbers," has been found satisfactory for making aldehyde-free 95 percent alcohol and carbonate-free potassium hydroxide used in making the di-(2-ethylhexyl)-sebacate and di-(2-ethylhexyl)-adipate determinations (see 4.4.3.2 and 4.4.3.3).

6.4 A Fisher Moroney "No-Bump" evaporator has been found satisfactory for making the di-(2-ethylhexyl)-sebacate and di-(2-ethylhexyl)-adipate determinations (see 4.4.3.2 and 4.4.3.3).

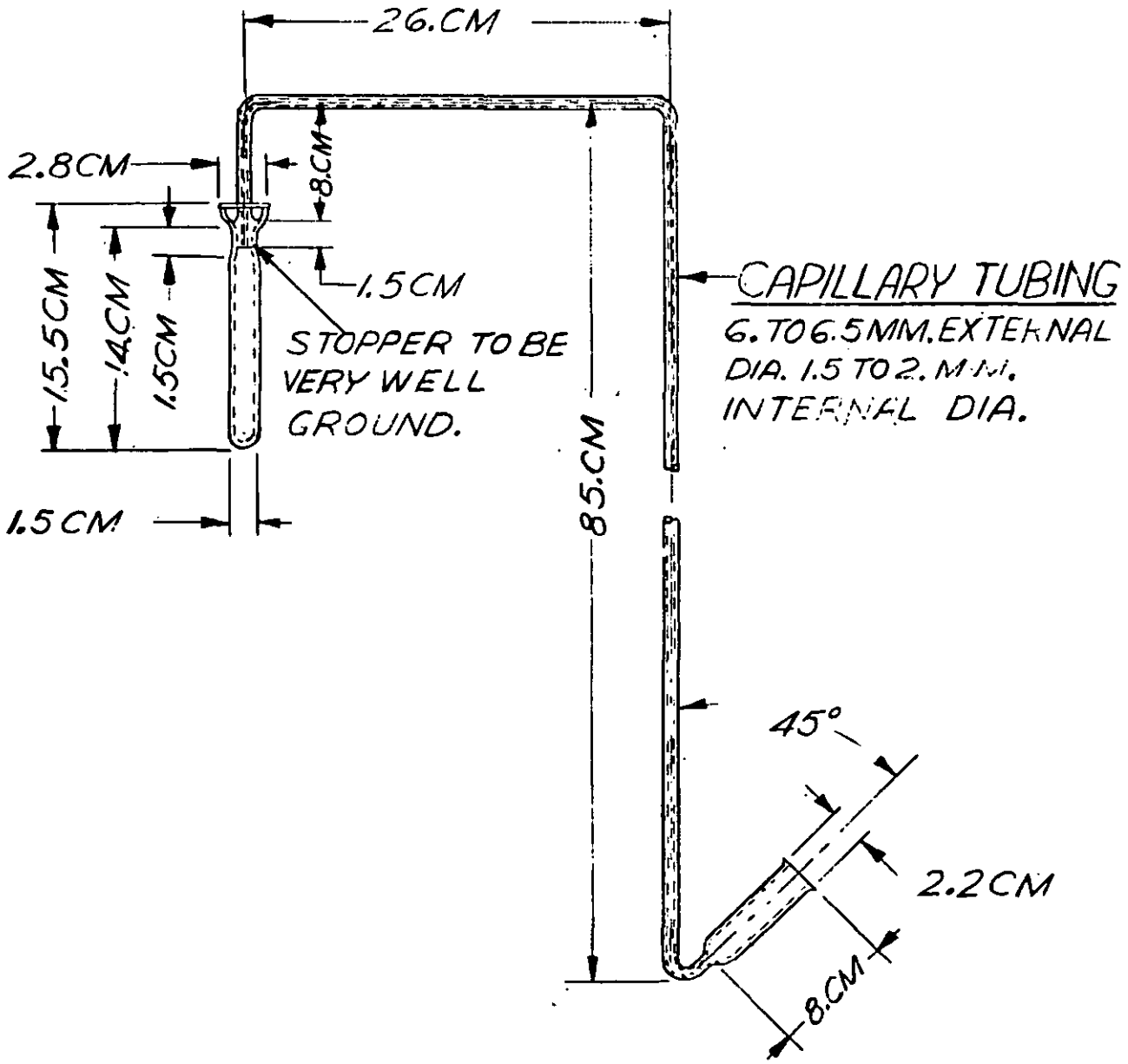
6.5 Stencilboard conforming to Specification UU-P-543, Type 2, Grade I, has been found satisfactory for making the tray used in the test for unincorporated lumps (see 4.4.7).

NOTICE. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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APPARATUS FOR
100°C. VACUUM STABILITY TEST

FIGURE - 1

GPO 926897

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